1(a)	$(-2, -3)$	B1
		(1)
(b)	$(-3, -9)$	B1B1
		(2)
(c)	$(-4, 3)$	B1
		(1)
		(4 marks)

2(a)	$\left(\mathrm{f}(2)=\right)2^4-5(2)^2+4(2)-7=-3<0$ **and** $\left(\mathrm{f}(3)=\right)3^4-5(3)^2+4(3)-7=41>0$	M1
	There is a change of sign and f(x) is continuous so there is a root (in the interval) *	A1*
		(2)
(b)	$x^3=\dfrac{5x^2-4x+7}{x}\Rightarrow x=\sqrt[3]{\dfrac{5x^2-4x+7}{x}}$ *	B1*
		(1)
(c)(i)	$x_2=\sqrt[3]{\dfrac{5(2)^2-4(2)+7}{2}}$	M1
	= awrt 2.1179	A1
(ii)	$\alpha=$ awrt 2.1565	A1
		(3)
		(6 marks)

3(a)	$\log_{10} D = 1.04 + 0.38t \Rightarrow D = 10^{1.04+0.38t}$ **or** $a = 10^{1.04}$ or $b = 10^{0.38}$	M1
	$a =$ awrt 10.96 **or** $b =$ awrt 2.399	A1
	$D = 10.96 \times 2.399^t$	A1
		(3)
(b)	$45000 = "10.96" \times "2.399"^T \Rightarrow T = ...$ or $\log_{10} 45000 = 1.04 + 0.38T \Rightarrow T = ...$	M1
	awrt 9.51	A1
		(2)
(c)	$D = "10.96" \times "2.399"^{12} \Rightarrow D = ...$ or $\log_{10} D = 1.04 + 0.38 \times 12 \Rightarrow D = ...$ or $350000 = "10.96" \times "2.399"^t \Rightarrow t = ...$ or $\log_{10} 350000 = 1.04 + 0.38 \times t \Rightarrow t = ...$	M1
	$D =$ awrt (£)400 000 $\Rightarrow$ yes or $t =$ awrt 11.9 $\Rightarrow$ yes	A1
		(2)
		(7 marks)

4(a)	$\mathrm{f}(x)=\dfrac{2x^2-32}{(3x-5)(x+4)}+\dfrac{8}{3x-5}$	B1
	$=\dfrac{2x^2-32}{(3x-5)(x+4)}+\dfrac{8}{3x-5}=\dfrac{2x^2-32+8(x+4)}{(3x-5)(x+4)}$ or e.g. $=\dfrac{2(x-4)(x+4)}{(3x-5)(x+4)}+\dfrac{8}{3x-5}=\dfrac{2(x-4)+8}{(3x-5)}$	M1
	$=\dfrac{2x\cancel{(x+4)}}{(3x-5)\cancel{(x+4)}}=\dfrac{2x}{3x-5}$ *	A1*
		(3)
(b)	$\mathrm{f}'(x)=\dfrac{2(3x-5)-3\times 2x}{(3x-5)^2}$	M1A1
	$\mathrm{f}'(x)=\dfrac{-10}{(3x-5)^2}$ As $(3x-5)^2>0$ (for $x>2$) then $\mathrm{f}'(x)<0$ Hence f is a decreasing function *	A1cso*
		(3)
(c)	$\mathrm{g}^{-1}(x)=\mathrm{e}^{\frac{x-3}{2}}$	M1A1
	$x \ldots 3$	B1
		(3)
(d)	$\mathrm{g}\left(\dfrac{2a}{3a-5}\right)=5\Rightarrow 3+2\ln\left(\dfrac{2a}{3a-5}\right)=5$	B1
	$\ln\left(\dfrac{2a}{3a-5}\right)=1\Rightarrow\dfrac{2a}{3a-5}=\mathrm{e}$	M1
	$\dfrac{2a}{3a-5}=\mathrm{e}\Rightarrow a=\ldots$	**d**M1
	$a=\dfrac{5\mathrm{e}}{3\mathrm{e}-2}$	A1
		(4)
(d) **Way 2**	$\mathrm{gf}(a)=5\Rightarrow\dfrac{2a}{3a-5}=\mathrm{g}^{-1}(5)$	B1
	$\dfrac{2a}{3a-5}=\mathrm{g}^{-1}(5)\Rightarrow\dfrac{2a}{3a-5}=\mathrm{e}$	M1
	$\dfrac{2a}{3a-5}=\mathrm{e}\Rightarrow a=\ldots$	**d**M1
	$a=\dfrac{5\mathrm{e}}{3\mathrm{e}-2}$	A1
		(13 marks)

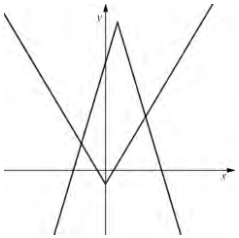
5(a)	$(A=)8$	B1
		(1)
(b)	$16=10+"8"\mathrm{e}^{-Bt}\Rightarrow \mathrm{e}^{-Bt}=...$	M1
	$\mathrm{e}^{-45B}="\frac{3}{4}"\Rightarrow -45B=\ln"\frac{3}{4}"\Rightarrow B=...$	**d**M1
	$B=\text{awrt}\,0.00639$	A1
		(3)
(c)	$\frac{\mathrm{d}T}{\mathrm{d}t}=-"8"\times\frac{1}{45}\ln\left(\frac{4}{3}\right)\times\mathrm{e}^{-\frac{1}{45}\ln\left(\frac{4}{3}\right)\times 2}=...$	M1
	$=-0.0505$	A1
		(2)
(d)	The temperature has a (lower) limit of 10 °C **or** $5=10+"8"\mathrm{e}^{-Bt}\Rightarrow \mathrm{e}^{-Bt}=-\frac{5}{"8"}$ e.g. which is not possible or cannot be solved or you cannot find the log of a negative number	B1
		(1)
		(7 marks)

6(a)	$\left(\frac{3\pi}{2}, 0\right)$	B1
		(1)
(b)	$f'(x) = \left(6\cos^2 x\right)e^{3\sin x} - \left(2\sin x\right)e^{3\sin x}$	M1A1
	$f'(x) = \left(6\cos^2 x\right)e^{3\sin x} - \left(2\sin x\right)e^{3\sin x}$ $\Rightarrow \left(6\left(1-\sin^2 x\right) - 2\sin x\right)(=0)$	**d**M1
	$\Rightarrow 3\sin^2 x + \sin x - 3 = 0$ oe	A1
		(4)
(c)	$\sin x = \frac{-1+\sqrt{37}}{6} \Rightarrow x = ...$	M1
	$x = 2.131$	A1
		(2)
		(7 marks)

7(a)	$\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right)-\frac{16}{3}(3x-k)^{-2}$	M1A1
		(2)
(b)	$-\frac{16}{3}(3x-k)^{-2}=-12\Rightarrow(3x-k)^2=...$	M1
	$3-k="\pm\frac{2}{3}"\Rightarrow k=...$	**d**M1
	$k=\frac{7}{3}\ ,\ \frac{11}{3}$	A1
		(3)
(c)	$y=\frac{16}{9\left(3(1)-\frac{7}{3}\right)}$	M1 (B1 on ePEN)
	$y-"\frac{8}{3}"=\frac{1}{12}(x-1)$	**d**M1
	$12y-x-31=0$	A1
		(3)
(d)	$\int\frac{16}{9(3x-k)}\mathrm{d}x=\frac{16}{27}\left[\ln(3x-k)\right]$	M1
	$=\frac{16}{27}\left[\ln\left(3x-\frac{7}{3}\right)\right]$	A1ft
	$\frac{16}{27}\left[\ln\left(3x-\frac{7}{3}\right)\right]_1^3=\frac{16}{27}\left(\ln\left(3(3)-\frac{7}{3}\right)-\ln\left(3(1)-\frac{7}{3}\right)\right)$	**d**M1
	$=\frac{16}{27}\ln(10)$	A1
		(4)
		(12 marks)

7(a)	$y=\frac{16}{9(3x-k)}=\frac{16}{27x-9k}\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{432}{(27x-9k)^2}$	M1A0
(b)	$-\frac{432}{(27x-9k)^2}=-12\Rightarrow 12(27x-9k)^2=432$	M1
	$12(27x-9k)^2=432\Rightarrow(27x-9k)^2=36\Rightarrow 27-9k=\pm 6\Rightarrow k=...$	**d**M1
	$k=\frac{7}{3}\ ,\ \frac{11}{3}$	A1
(c)	$y=\frac{16}{27-21}$	M1 (B1 on ePEN)
	$y-"\frac{8}{3}"=\frac{1}{12}(x-1)$	**d**M1
	$12y-x-31=0$	A1
(d)	$\int\frac{16}{27x-9k}\mathrm{d}x=\frac{16}{27}\left[\ln(27x-9k)\right]$	M1
	$=\frac{16}{27}\left[\ln(27x-21)\right]$	A1ft
	$\frac{16}{27}\left[\ln(27x-21)\right]_1^3=\frac{16}{27}\left(\ln(27(3)-21)-\ln(27-21)\right)$	**d**M1
	$=\frac{16}{27}\ln(10)$	A1

8(a)(i)	$\left(\frac{b}{2}, a\right)$	B1B1
(ii)	$(0, a-b)$	B1
(iii)	$\left(\frac{b-a}{2}, 0\right)$ and $\left(\frac{a+b}{2}, 0\right)$	B1B1
		(5)

(b)		B1B1
		(2)
(c)	$-x-1=2x+a-b,\ x=-3\Rightarrow 2=-6+a-b$ or $x-1=a+b-2x,\ x=5\Rightarrow 5-1=a+b-10$	M1
	$-x-1=2x+a-b,\ x=-3\Rightarrow 2=-6+a-b$ and $x-1=a+b-2x,\ x=5\Rightarrow 5-1=a+b-10$	dM1 (A1 on ePEN)
	$\begin{matrix}a-b=8\\a+b=14\end{matrix}\Rightarrow a=...$ or $b=...$	**dd**M1
	$a=11,\ b=3$	A1
		(4)
		(11 marks)

9(a)	$\dfrac{3\sin\theta\cos\theta}{\cos\theta+\sin\theta}=(2+\sec 2\theta)(\cos\theta-\sin\theta)$ $\Rightarrow \dfrac{3}{2}\sin 2\theta=(2+\sec 2\theta)(\cos\theta-\sin\theta)\left(\cos\theta+\sin\theta\right)$ **or** $\Rightarrow 3\sin\theta\cos\theta=(2+\sec 2\theta)\cos 2\theta$	M1
	$\Rightarrow \dfrac{3}{2}\sin 2\theta=(2+\sec 2\theta)\cos 2\theta$	**d**M1
	$\Rightarrow \dfrac{3}{2}\sin 2\theta=(2+\sec 2\theta)\cos 2\theta$ $\Rightarrow \dfrac{3}{2}\sin 2\theta=2\cos 2\theta+1\Rightarrow 3\sin 2\theta-4\cos 2\theta=2$ *	A1*
		(3)

(b)	Way 1 using $R\sin(2x-\alpha)$ or $R\cos(2x+\alpha)$	
	$\Rightarrow R=\sqrt{3^2+4^2}=...(5)$ **or** $(\alpha=)\ \tan^{-1}\left(\frac{4}{3}\text{ or }\frac{3}{4}\right)=...$	M1
	$\Rightarrow R=\sqrt{3^2+4^2}=...(5)$ **and** $(\alpha=)\ \tan^{-1}\left(\frac{4}{3}\text{ or }\frac{3}{4}\right)=...$	**d**M1
	$3\sin 2x-4\cos 2x=2\Rightarrow 5\sin(2x-0.927)=2$ **or** $3\sin 2x-4\cos 2x=2\Rightarrow 5\cos(2x+0.644)=-2$	A1
	$5\sin(2x-0.927)=2\Rightarrow x=\dfrac{\sin^{-1}\frac{2}{5}+0.927}{2}$ **or** $5\cos(2x+0.644)=-2\Rightarrow x=\dfrac{\cos^{-1}\frac{-2}{5}-0.644}{2}$	**dd**M1
	$x=$ awrt 3.81	A1
		(5)
		(8 marks)

(b)	Way 2 Squaring	
	$3\sin 2x - 4\cos 2x = 2 \Rightarrow 3\sin 2x = 2 + 4\cos 2x$ $\Rightarrow 9\sin^2 2x = 4 + 16\cos 2x + 16\cos^2 2x$ or $3\sin 2x - 4\cos 2x = 2 \Rightarrow 3\sin 2x - 2 = 4\cos 2x$ $\Rightarrow 9\sin^2 2x - 12\sin 2x + 4 = 16\cos^2 2x$	M1
	$9\left(1 - \cos^2 2x\right) = 4 + 16\cos 2x + 16\cos^2 2x$ $\Rightarrow 25\cos^2 2x + 16\cos 2x - 5 = 0$ or $9\sin^2 2x - 12\sin 2x + 4 = 16\left(1 - \sin^2 2x\right)$ $\Rightarrow 25\sin^2 2x - 12\sin 2x - 12 = 0$	**d**M1A1
	$25\cos^2 2x + 16\cos 2x - 5 = 0 \Rightarrow \cos 2x = \frac{-8 \pm 3\sqrt{21}}{25}\left(0.23,\ -0.87\right)$ or $25\sin^2 2x - 12\sin 2x - 12 = 0 \Rightarrow \sin 2x = \frac{6 \pm 4\sqrt{21}}{25}\left(0.97,\ -0.49\right)$ $\Rightarrow x = \ldots$	**dd**M1
	$x =$ awrt 3.81	A1

(b)	Way 3 Using double angle formulae	
	$3\sin 2x - 4\cos 2x = 2 \Rightarrow 6\sin x\cos x - 4\cos^2 x + 4\sin^2 x = 2$	M1
	$6\sin x\cos x - 4\cos^2 x + 4\sin^2 x = 2 \Rightarrow 6\tan x - 4 + 4\tan^2 x = 2\sec^2 x$ $\Rightarrow 6\tan x - 4 + 4\tan^2 x = 2\left(1 + \tan^2 x\right)$ $\Rightarrow \tan^2 x + 3\tan x - 3 = 0$	**d**M1A1
	$\Rightarrow \tan^2 x + 3\tan x - 3 = 0 \Rightarrow \tan x = \dfrac{-3 \pm \sqrt{21}}{2}\left(0.79, -3.79\right)$ $\Rightarrow x = ...$	**dd**M1
	$x =$ awrt 3.81	A1

(b)	Way 4 Divides by cos2x and squares	
	$3\sin 2x - 4\cos 2x = 2 \Rightarrow 3\tan 2x - 4 = 2\sec 2x$ $\Rightarrow 9\tan^2 2x - 24\tan 2x + 16 = 4\sec^2 2x$	M1
	$\Rightarrow 9\tan^2 2x - 24\tan 2x + 16 = 4\left(1 + \tan^2 2x\right)$ $\Rightarrow 5\tan^2 2x - 24\tan 2x + 12 = 0$	**d**M1A1
	$5\tan^2 2x - 24\tan 2x + 12 = 0 \Rightarrow \tan 2x = \frac{12 \pm 2\sqrt{21}}{5}\left(4.23,\ 0.57\right)$ $\Rightarrow x = ...$	**dd**M1
	$x =$ awrt 3.81	A1